

Compound Inequalities

When two statements are joined, they form a **compound statement**.

Definitions

A **conjunction** is a compound statement that uses the word *and*.

$$x < 0 \text{ and } x \geq -3$$

A **disjunction** is a compound statement that uses the word *or*.

$$x = 0 \text{ or } x = 5 \quad x > 0 \text{ or } x < 23$$

A **compound inequality** is a pair of inequalities joined by the word *and* or the word *or*. A compound inequality is a type of compound statement.

Conjunction and Inequalities

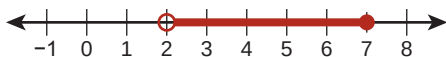
The solution set of a conjunction inequality is the set of all numbers that satisfy both inequalities.

EXAMPLE 1

Graph the compound inequality, and then express it in interval notation.

A $x > 2$ and $x \leq 7$ (also written $2 < x \leq 7$)

SOLUTION

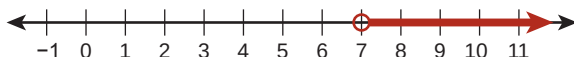


This set is of all numbers between 2 and 7, not including 2 but including 7.

$$x \in (2, 7]$$

B $x > 5$ and $x > 7$

SOLUTION

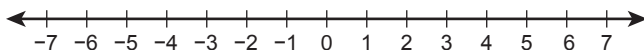


If a number is both greater than 5 and greater than 7, then it must be greater than 7.

$$x \in (7, \infty)$$

C $x > 0$ and $x < -2$

SOLUTION



There is no number that is both greater than 0 and less than -2 .

There is no interval. The solution set is $\emptyset$, the empty set. ■

The four general cases for conjunction inequalities, such as the one in Example 1A, are shown in the table.

Inequality notation	Graph	Interval notation
$a < x < b$		$x \in (a, b)$
$a \leq x \leq b$		$x \in [a, b]$
$a < x \leq b$		$x \in (a, b]$
$a \leq x < b$		$x \in [a, b)$

► **Remember** An open interval has neither endpoint included, a closed interval has both endpoints included, and a half-open interval (also called a half-closed interval) has one endpoint included.

Disjunction and Inequalities

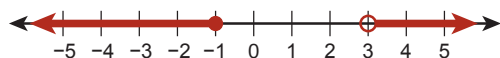
The solution set of a disjunction inequality is the set of all numbers that satisfy either or both inequalities.

EXAMPLE 2

Graph the compound inequality, and then express it in interval notation.

A $x \leq -1$ or $x > 3$

SOLUTION



This set is of all numbers that are either less than or equal to -1 or greater than 3 .

$$x \in (-\infty, -1] \cup x \in (3, \infty)$$

B $t > 4$ or $t > 9$

SOLUTION

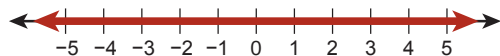


The set of all numbers that are either greater than 4 or greater than 9 is just the set of all numbers that are greater than 4 .

$$t \in (4, \infty)$$

C $z > -2$ or $z < 1$

SOLUTION



The set of all numbers that are either greater than -2 or less than 1 is the set of all real numbers.

$$z \in (-\infty, \infty) \blacksquare$$

Writing a Compound Inequality and Interval Notation for a Given Graph

EXAMPLE 3

For the graph, write a compound inequality and interval notation.

A



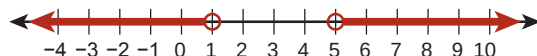
SOLUTION

This set is of numbers between -3 and 3 , excluding -3 but including 3 .

$$\text{inequality: } -3 < x \leq 3$$

$$\text{interval notation: } x \in (-3, 3]$$

B



SOLUTION

This set is of numbers that are either less than 1 or greater than 5 .

$$\text{inequality: } x < 1 \text{ or } x > 5$$

$$\text{interval notation: } x \in (-\infty, 1) \cup (5, \infty) \quad \blacksquare$$

Solving and Graphing a Compound Inequality in One Variable

EXAMPLE 4

Solve the compound inequality. Graph the solution set.

A $5 + x < 7$ or $-2x \leq -10$

SOLUTION

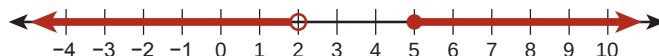
Solve each inequality separately.

First Inequality

$$5 + x < 7 \text{ or } -2x \leq -10$$

Subtract 5 from
both sides.

$$x < 2 \text{ or } x \geq 5$$

Second InequalityDivide both sides
by -2 , reversing the
inequality sign.

► **Remember** To solve an inequality in one variable, find all values for the variable that make the inequality a true statement.

B $3x - 1 \geq 14$ and $6x \leq 48$

SOLUTION

Solve each inequality separately.

$$3x - 1 \geq 14 \text{ and } 6x \leq 48$$

Add 1 to both sides.

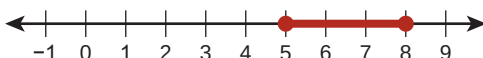
$$3x \geq 15 \text{ and } x \leq 8$$

Divide both sides

Divide both sides by 3.

$$x \geq 5 \text{ and } x \leq 8$$

by 6.



► **Think About It** The compound inequality $x \geq 5$ and $x \leq 8$ can be written as $5 \leq x \leq 8$.

C $4 \leq 2n + 10 \leq 18$

SOLUTION

$$4 \leq 2n + 10 \leq 18$$

Original inequality

$$-6 \leq 2n \leq 8$$

Subtract 10 from all expressions.

$$-3 \leq n \leq 4$$

Divide all expressions by 2.

